BALAD FIELD · DRILLING ENGINEERING

Unified Pore Pressure and Fracture Gradient (PPFG)

Method, calculations and results from nine offset wells (Balad-1 to Balad-9)

Input	Balad_PPFG__PT.xlsx – Final PPFG sheets of Balad-1 … Balad-9, EBS-1996 study table, Pressure Testing (casing) sheet
Outputs	Balad_Unified_PPFG.xlsx, Balad_PPFG_Final.png, Balad_PPFG_Final_Wells.png, Balad_PPFG_QC_AllWells.png, Balad_OBG_QC.png
Code	build_unified_ppfg.py (all numbers in this report are produced by it)
Date	17 October 2026

Summary

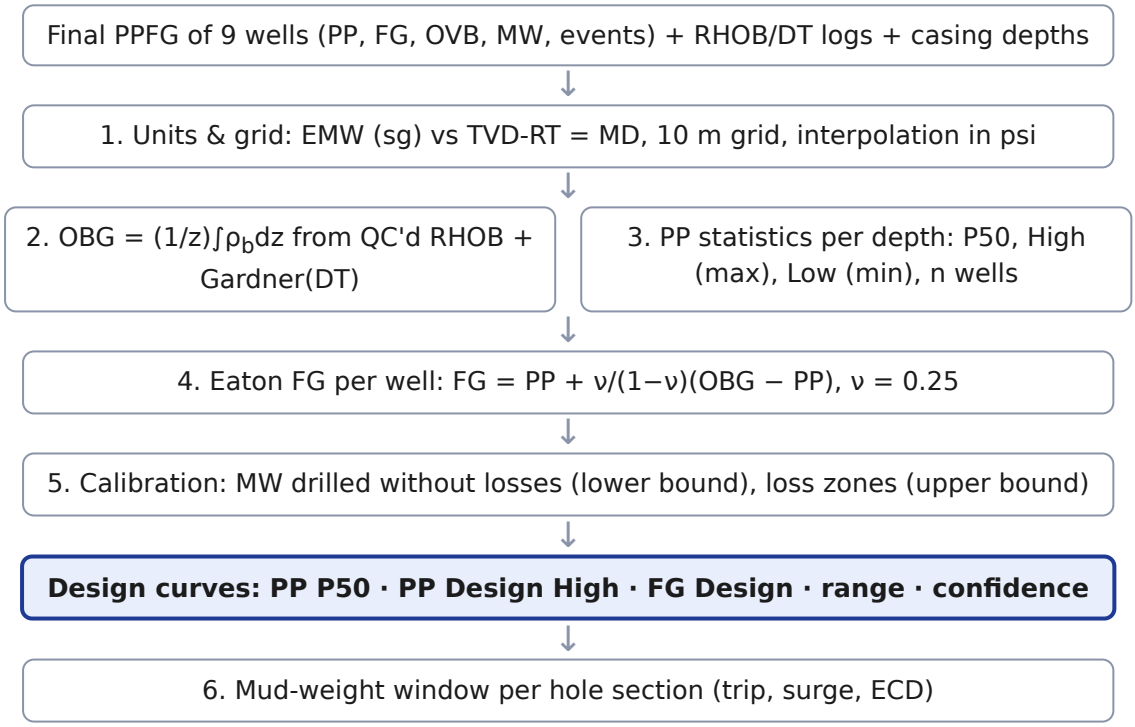
The nine wells' interpreted pore pressure (PP) tables were put on one unit (EMW, sg) and one 10 m depth grid. A new field overburden gradient (OBG) was integrated from QC'd density logs, replacing the workbook's constant-density OVB. At every depth the wells were combined into **PP P50** (median), **PP Design High** (maximum of the wells) and a min–max range, with the number of supporting wells shown as a confidence measure.

Fracture gradient was computed for each well with Eaton's equation ($\nu = 0.25$) and the field OBG, then calibrated with the drilling record because no LOT/FIT exist: mud weights drilled without losses raise a well's FG (lower bound), and loss events cap it (upper bound). **FG Design** is Eaton applied to PP Design High – the same pressure case, so the two design curves never cross – capped by the loss zones (0–53 m: 1.10 sg; 2262–2305 m: 1.23 sg; 2610–3210 m: 1.18 sg). Finally the mud-weight window was computed for each hole section with trip, surge and ECD allowances.

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Method flow



1. Input data

All inputs come from **Balad_PPFG__PT.xlsx**. Each of the nine sheets *Balad-N Final PPFG* holds the user's own final interpretation for that well; this method takes those values as given and does not re-interpret logs for PP.

1.1 What was read from each well sheet

- **Final PPFG table** (starting at row 10): Depth (m), PP (psi), OVB (psi), FP = fracture pressure (psi), MW (FWR) = mud weight actually used (sg).
- **Event notes** in the column left of *Depth*, and kill/resume mud weights to the right of the table. Notes are classified by keyword: “kick” → **Kick**; “loss” → **Losses**; “gas cut” or “dropped” → **Gas-cut mud**. TD notes are ignored.
- **Raw logs**: DT (µs/ft) in the first depth block, RHOB (g/cc) and DRHO (g/cc) in the second depth block, 0.25 m sampling.

1.2 Coverage per well

Well	TD (m)	PPFG rows	DT kept (m)	RHOB kept (m)	Kicks	Losses	Gas-cut
Balad-1	4432	58	520–4270	1250–3590	7	1	0
Balad-2	4360	56	270–4314	1300–3614	2	3	0
Balad-3	4260	59	272–4314	1600–4193	1	3	5
Balad-4	3140	43	366–3150	1877–3150	0	5	0
Balad-5	2480	30	870–1880	1840–2490	0	0	0
Balad-6	2183	27	865–2184	1815–2185	0	1	0
Balad-7	2448	30	1120–2455	1829–2454	0	0	0
Balad-8	3312	45	209–3317	1875–3319	0	5	0
Balad-9	2480	33	870–3317	1872–3319	0	0	0

1.3 Other sheets used

- **EBS PPFG** – the 1996 East Baghdad study table (OBG, PP and FG in psi/ft every 100 m to 3500 m). Used only as a reference.
- **Pressure Testing** – casing float-collar depths (one well), used as the hole-section depths in Section 6.
- Formation tops available: Hartha/Sa'adi 1980 m and Khasib 2325 m (Balad-1 sheet / 1996 study).

1.4 Not available

No LOT/FIT, no MDT/RFT pressures, no RT/GL elevations, no complete formation tops and no per-well casing depths. These are old wells; the consequences are handled in Sections 5–6 and listed in Section 8.

2. Units, datum and the common depth grid

2.1 Constants (same as the workbook)

Constant	Value	Note
Pressure gradient per ppg	0.0519 psi/ft/ppg	workbook factor
ppg per sg	8.337	workbook factor
ft per m	3.281	workbook factor
psi/ft per sg	0.43269	0.0519 × 8.337
psi/m per sg	1.41966	0.43269 × 3.281

2.2 Conversion to equivalent mud weight

$$\text{EMW (sg)} = P\ (\text{psi}) \,/\, [\ D\ (\text{m}) \times 1.4197\ ]$$

Depth reference: the wells are vertical, so TVD = MD, and every depth stays on the well's own rotary table (RT) as recorded. No RT elevations exist, so no TVDSS correction was applied.

Worked example – Balad-2 at 2000 m (row of the Final PPFG table):

Quantity	Pressure (psi)	Gradient (psi/ft)	EMW (sg)	EMW (ppg)
PP	3,343.4	0.5095	1.178	9.82
FP (fracture)	4,315.1	0.6576	1.520	12.67
OVB	6,258.6	0.9538	2.204	18.38

Example for PP: 3,343.4 / (2000 × 1.4197) = **1.178 sg**.

2.3 Common 10 m grid and interpolation in pressure

All wells are re-sampled to depths 0, 10, 20 … 4440 m so that the nine wells are compared at identical depths. Each well is used only between its first and last table depth. Interpolation is done on **pressure in psi**, then converted to EMW: EMW = P/D is not linear in depth, and the tables contain step changes at kicks.

Worked example – Balad-1 at 1950 m, between table rows 1900 m (3,101.3 psi) and 1980 m (3,310.0 psi):

$$P(1950) = 3{,}101.3 + (1950 - 1900) \,/\, (1980 - 1900) \times (3{,}310.0 - 3{,}101.3) = 3{,}231.7 \text{ psi} \rightarrow \text{EMW} = 1.1674 \text{ sg}$$

Interpolating the EMW values directly would give 1.1671 sg; the pressure route is the physically consistent one.

2.4 Normal hydrostatic and datum sensitivity

The reference line is the normal hydrostatic gradient 0.465 psi/ft = **1.075 sg**, measured from RT. A datum error ΔD changes EMW by the factor D/(D+ΔD): a 10 m error at 1000 m changes 1.18 sg by 0.012 sg (1%). The effect is negligible below a few hundred metres and largest near surface.

3. Overburden gradient (OBG)

The overburden stress is the weight of all rock and fluid above a depth. The workbook's OVB used a constant (or point) bulk density times depth, which gives 2.35–2.54 sg from the surface and jumps with every density spike. Because FG depends directly on OBG, a new field OBG was integrated from the density logs.

$$\sigma_v(z) = g \int_0^z \rho_b(z')\, dz' \quad \Rightarrow \quad \text{OBG (sg)} = (1/z) \int_0^z \rho_b\, dz \quad (\rho_b \text{ in g/cc})$$

3.1 Log QC

- RHOB kept only if 1.6 < RHOB < 3.05 g/cc and |DRHO| ≤ 0.15 g/cc (bad-hole / poor pad contact).
- DT kept only if 35 < DT < 200 µs/ft. Above 1200 m, DT < 52 µs/ft is rejected: values of 48–50 µs/ft at the top of Balad-3, Balad-4 and Balad-8 are readings inside casing, not formation. The first 20 m of each DT log is also dropped.

Well	DT raw	DT out of range	DT in casing	DT kept*	RHOB raw	RHOB out of range	|DRHO|>0.15	RHOB kept
Balad-1	15,081	4	52	14,944	9,361	0	0	9,361
Balad-2	7,254	2	36	7,135	9,257	14	0	9,243
Balad-3	7,254	916	585	5,672	9,251	171	468	8,612
Balad-4	6,814	0	389	6,356	6,202	0	28	6,174
Balad-5	4,121	56	262	3,769	2,599	0	2	2,597
Balad-6	5,357	0	290	4,986	1,481	0	1	1,480
Balad-7	5,420	0	12	5,327	2,559	0	56	2,503
Balad-8	12,389	0	396	11,982	5,817	0	40	5,777
Balad-9	12,139	129	2	11,939	5,220	0	68	5,152

* DT kept also excludes the first 20 m of each log.

3.2 Density where RHOB is missing – Gardner calibrated on Balad

RHOB starts at 1250–1875 m, but DT starts at 180–1100 m. Above the RHOB log, density is estimated from DT with Gardner et al. (1974):

$$\rho_b = a \cdot V^{0.25}, \quad V = 10^6 / \text{DT (ft/s)}$$

The coefficient was calibrated on the 10 m bins where both RHOB and DT exist (676 bins over all wells): a = median(RHOB / $V^{0.25}$) = **0.2250** (textbook value 0.23).

3.3 Field composite density

1. Each well: 10 m bin medians of RHOB; Gardner density where RHOB is missing.
2. Field density = median of the wells in each bin, used only where at least 2 wells have data.
3. Gaps filled linearly; 50 m running-median smoothing.
4. Above the first supported bin (270 m) a linear trend from 1.95 g/cc at surface (assumed value for unconsolidated near-surface sediments).

3.4 Integration – worked example to 1000 m

The integral is a cumulative trapezoid sum on the 10 m grid. The table groups it in 250 m intervals:

Interval (m)	Mean density (g/cc)	∫ρdz in interval (g/cc·m)	Cumulative ∫ρdz	OBG at base (sg)
0–250	2.105	526.2	526.2	2.105
250–500	2.295	573.9	1,100.1	2.200
500–750	2.255	563.7	1,663.9	2.218

Interval (m)	Mean density (g/cc)	∫ρdz in interval (g/cc·m)	Cumulative ∫ρdz	OBG at base (sg)
750–1000	2.241	560.2	2,224.1	2.224

OBG(1000 m) = 2,224.1 / 1000 = **2.224 sg** = 0.962 psi/ft.

3.5 Comparison with the 1996 study and the workbook OVB

Depth (m)	Field density (g/cc)	Field OBG (sg)	EBS-1996 OBG (sg)	Difference (sg)	Workbook OVB range (sg)
100	2.074	2.013	2.103	-0.090	2.35–2.54
500	2.310	2.200	2.184	0.016	2.35–2.54
1000	2.372	2.224	2.242	-0.018	2.35–2.54
1500	2.419	2.300	2.311	-0.011	2.35–2.54
2000	2.300	2.322	2.369	-0.047	2.20–2.54
2500	2.540	2.349	2.392	-0.043	2.40–2.73
3000	2.690	2.391	2.415	-0.024	1.96–2.80
3500	2.340	2.409	2.427	-0.017	2.31–2.40

Over the EBS-1996 depths the field OBG differs by -0.024 sg on average (maximum |difference| 0.090 sg, at shallow depth where the surface trend is assumed). The workbook OVB scatters from 1.95 to 2.93 sg, which is why it was replaced.

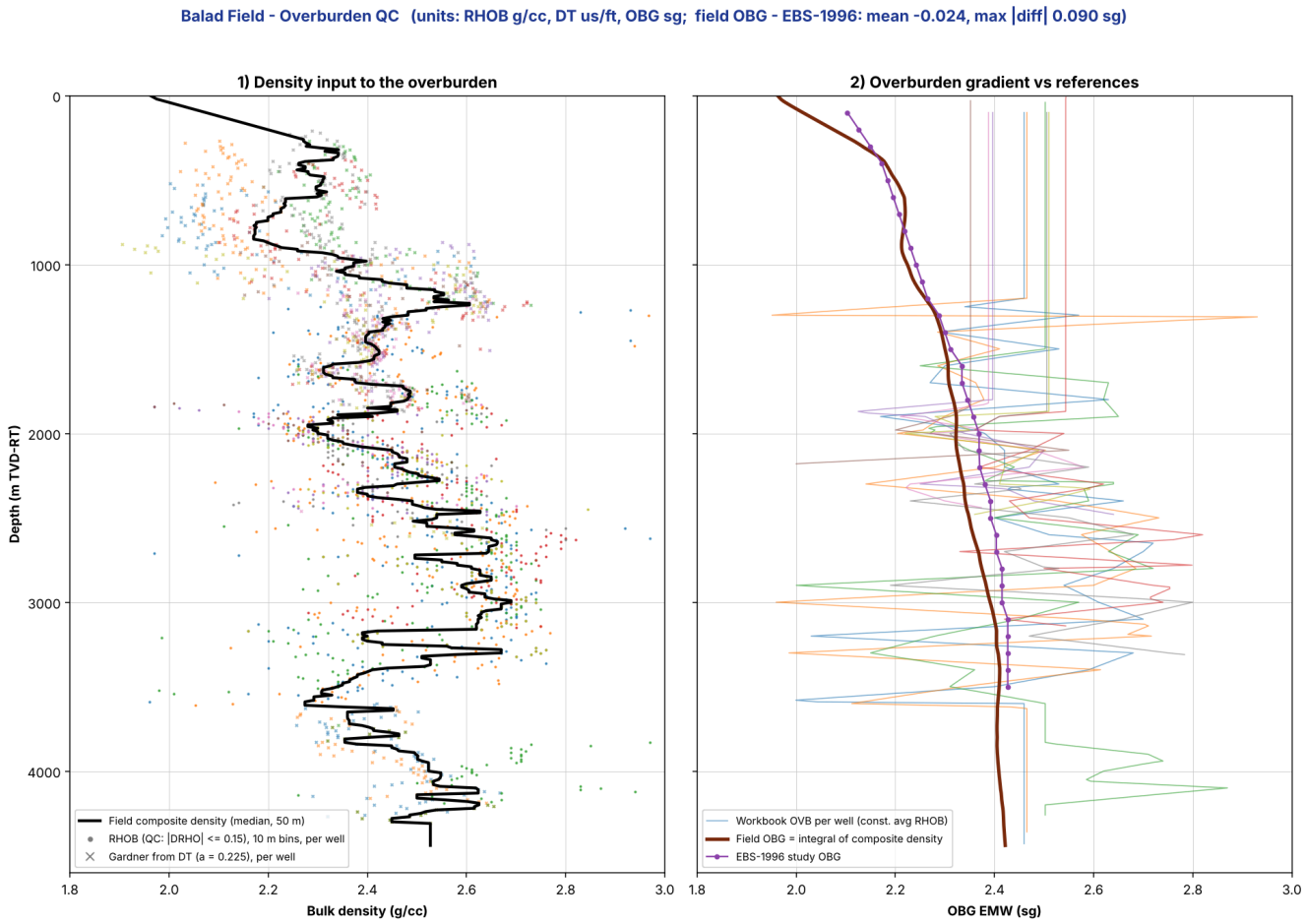


Figure 1 – OBG QC: density inputs per well and field composite (left); field OBG vs EBS-1996 and workbook OVB (right).

4. Unified pore pressure and number of wells

At every grid depth the wells that have data there are combined (well count n):

- **PP Low** = minimum of the wells.
- **PP P50** (most likely) = median of the wells, then a 5-point (50 m) running median to remove single-sample noise.
- **PP Design High** = maximum of the wells (the offset envelope), not smoothed. It is used for kick tolerance, burst loads and the minimum mud weight, because a new well can meet the highest pressure seen in any offset (Bourgoyne et al., 1986; Zhang, 2011).

4.1 Worked example – each well's PP (sg) at selected depths

Depth (m)	Ba-1	Ba-2	Ba-3	Ba-4	Ba-5	Ba-6	Ba-7	Ba-8	Ba-9	n	Low	P50	High
1000	1.12	1.18	1.18	1.14	1.18	1.18	1.18	1.18	1.18	9	1.12	1.18	1.18
2000	1.16	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	9	1.16	1.18	1.18
3000	1.16	1.16	1.16	1.16	–	–	–	1.16	–	5	1.16	1.16	1.16
3800	1.21	1.43	1.16	–	–	–	–	–	–	3	1.16	1.21	1.43
4000	1.27	1.86	1.68	–	–	–	–	–	–	3	1.27	1.77	1.86
4100	1.85	2.02	2.19	–	–	–	–	–	–	3	1.85	2.02	2.19

4.2 Confidence from the number of wells

Depth interval (m)	Confidence	Wells
10–30	Low (≤ 2 wells)	1–2
40–90	Medium (3–5 wells)	3–3
100–2480	High (≥ 6 wells)	7–9
2490–4260	Medium (3–5 wells)	3–5
4270–4430	Low (≤ 2 wells)	1–2

4.3 Transition to overpressure

Only Balad-1, Balad-2 and Balad-3 reach the deep overpressure. The first depth where each well's PP exceeds 1.3 sg differs by several hundred metres, so the High envelope rises earlier than any single well. This spread is the main reason for low confidence in the deep section:

Well	First depth with PP > 1.3 sg (m)	Maximum PP (sg)
Balad-1	4010	2.20
Balad-2	3690	2.03
Balad-3	3840	2.20

4.4 Check against kicks

A kick means PP exceeded the mud weight in use; the kill weight is an approximate upper bound. The unified curves at the kick depths:

Well	Depth (m)	MW at kick (sg)	Kill / max MW (sg)	PP P50 (sg)	PP Design High (sg)
Balad-1	3588	1.19	1.23	1.16	1.21
Balad-1	3991	1.23	1.95	1.76	1.83
Balad-1	4020	1.78	1.93	1.86	1.96
Balad-1	4139	1.93	2.17	2.01	2.18
Balad-1	4171	2.11	2.25	2.13	2.17
Balad-1	4206	2.17	2.25	2.16	2.20
Balad-1	4271	2.10	–	2.04	2.10

Well	Depth (m)	MW at kick (sg)	Kill / max MW (sg)	PP P50 (sg)	PP Design High (sg)
Balad-2	4000	1.78	1.88	1.77	1.86
Balad-2	4035	1.88	2.25	2.02	2.04
Balad-3	3940	2.00	2.22	1.66	2.05

4.5 Data features to note

- Some wells carry PP below hydrostatic near surface (e.g. 0.65 sg at 100 m) and others use 0.51 psi/ft (1.18 sg) from shallow depth, taken from the 1996 study. Both come from the input tables.
- Balad-2 has PP of about 0.99 sg at 3130–3200 m, inside the loss zone.

5. Fracture gradient and calibration

5.1 Eaton's equation

$$FG = PP + [\nu / (1 - \nu)] \cdot (OBG - PP), \quad \nu = 0.25 \Rightarrow \nu/(1-\nu) = 0.3333$$

Eaton (1969) treats the minimum horizontal stress as the effective vertical stress scaled by ν/(1−ν), plus pore pressure. ν = 0.25 is the value used in the user's workbook. FG is computed for **each well** with its own PP and the **field** OBG, so the OVB noise of the workbook does not reach FG.

Worked example at 2000 m (most-likely case): OBG = 2.322 sg, PP P50 = 1.178 sg

$$FG\ ML = 1.178 + 0.3333 \times (2.322 - 1.178) = \mathbf{1.559\ sg}$$

5.2 Why the EBS-1996 FG is not used for design

The 1996 study uses 2ν/(1−ν) = 0.667 instead of 0.333. That form is the **breakdown** pressure of an intact, unfractured borehole (Haimson & Fairhurst, 1967), which is higher than the fracture propagation/closure pressure that controls lost circulation. At 2000 m it would give 1.941 sg with the field OBG (EBS-1996 table: 1.964 sg), against 1.559 sg from Eaton.

5.3 Calibration without LOT/FIT

Lower bound – mud weight held without losses. If a well was drilled through depth z with mud weight MW and had no losses within ±20 m, the formation held MW, so that well's FG at z is at least MW (the use of offset drilling data in Daines, 1982). Where Eaton gives less, FG is raised to MW:

$$FG_{well}(z) = \max[\ FG_{Eaton}(z),\ MW_{no\ loss}(z)\ ]$$

Well	Grid points raised	Depth ranges (m)	Largest raise (sg)
Balad-1	1	4000–4000	0.13
Balad-2	28	4090–4360	0.14
Balad-3	18	3840–3910, 3950–4040	0.24

Upper bound – loss zones. Loss events of all wells are sorted by depth and grouped when consecutive events are less than 300 m apart. Each zone runs from 20 m above the first to 20 m below the last event, and its limit is the lowest mud weight at which losses occurred:

Loss zone (m)	Limit (sg)	Wells
0–53	1.10	Balad-3, Balad-4, Balad-6
2262–2305	1.23	Balad-3
2610–3210	1.18	Balad-1, Balad-2, Balad-4, Balad-8

Losses in the 2610–3210 m carbonates may be vugular or natural-fracture losses rather than induced fractures. Either way, they are a real ceiling for the mud weight in that interval.

5.4 FG Design – same pressure case as PP Design High

Within every single well PP is below FG at every depth (Eaton always gives FG > PP because OBG > PP). A crossover can only appear when the highest PP of one well is plotted against the lowest FG of another well at the same depth. That is what happens in the transition zone, because the overpressure top is at a different depth in each deep well. At 3940 m:

Well	PP (sg)	Calibrated FG (sg)
Balad-1	1.24	1.63
Balad-2	1.66	1.91
Balad-3	2.05	2.17

Taking PP from the overpressured well and FG from the still-normal well would put PP above FG by 0.42 sg – a state no well shows. The design FG is therefore taken from the **same pressure case** as PP Design High:

$$\text{FG Design}(z) = \min\left[\ \text{PP}_{\text{High}}(z) + \nu/(1-\nu)\cdot(\text{OBG}(z) - \text{PP}_{\text{High}}(z)),\quad \text{loss-zone limit}(z)\ \right]$$

FG Low (the weakest calibrated offset well) is kept as a dashed reference curve. It shows the risk in the transition zone: if a new well is still normally pressured where the design weight is already high, the formation may break down at the FG Low value. That is a casing-point and real-time pore-pressure-monitoring issue, not a curve to design the window on.

Depth (m)	OBG	PP P50	PP High	FG ML	FG Low (weakest well)	FG on PP High	Loss limit	FG Design	FG Design − PP High
500	2.20	1.11	1.18	1.47	1.45	1.52	–	**1.52**	+0.34
1000	2.22	1.18	1.18	1.53	1.49	1.53	–	**1.53**	+0.35
2000	2.32	1.18	1.18	1.56	1.55	1.56	–	**1.56**	+0.38
2800	2.37	1.16	1.16	1.56	1.56	1.56	1.18	**1.18**	+0.02
3500	2.41	1.16	1.16	1.57	1.57	1.57	–	**1.57**	+0.42
3800	2.41	1.21	1.43	1.61	1.57	1.76	–	**1.76**	+0.32
4000	2.41	1.77	1.86	1.99	1.78	2.04	–	**2.04**	+0.18
4200	2.42	2.14	2.16	2.23	2.23	2.25	–	**2.25**	+0.08

All values in sg. Note that Balad-1 and Balad-2 were drilled at about 2.25 sg below 4000 m; Eaton with ν = 0.25 may underestimate the deep FG, which only LOT/FIT data can confirm.

6. Mud-weight window per hole section

A mud weight must stay above pore pressure (with a margin for swab when tripping) and below the weakest fracture gradient in the open hole (with a margin for surge and for the extra pressure of circulation, ECD).

$$\text{Min MW} = \max_{\text{section}}(\text{PP Design High}) + 0.03 \text{ sg (trip / swab, } \approx 0.25 \text{ ppg)}$$

$$\text{Max static MW} = \min_{\text{open hole}}(\text{FG Design}) - 0.03 \text{ sg (surge)} - \Delta\text{ECD(hole size)}$$

ΔECD allowances are **assumed** typical values, not hydraulics results: 26″ 0.02, 17-1/2″ 0.03, 12-1/4″ 0.04, 8-1/2″ 0.06, 6″ 0.07 sg.

6.1 Hole sections from the casing depths

Casing	Float collar (ft)	Depth (m)
30"	39.4	12.0
20"	823.5	251.0
13-3/8"	4,183.3	1,275.0
9-5/8"	11,798.5	3,596.0
7"	14,223.1	4,335.0

6.2 Section results (sg)

Hole	Casing	Interval (m)	Wells	Max PP High @ m	Min FG Design @ m	ΔECD	Min MW	Max static MW	Window	Window excl. loss zones	Recommended MW	MW used in wells
26"	20"	12–251	1–9	1.18 @ 100	1.10 @ 20	0.02	1.21	1.05	-0.16	-0.11	**1.21**	1.06–1.25
17-1/2"	13-3/8"	251–1275	9–9	1.18 @ 350	1.49 @ 260	0.03	1.21	1.43	+0.22	+0.22	**1.21**	1.08–1.31
12-1/4"	9-5/8"	1275–3596	3–9	1.23 @ 2030	1.18 @ 2610	0.04	1.26	1.11	-0.15	+0.21	**1.26**	1.17–1.40
8-1/2"	7" liner	3596–4335	2–3	2.20 @ 4060	1.61 @ 3630	0.06	2.23	1.52	-0.71	-0.71	**2.23**	1.21–2.25
6"	open hole to TD	4335–4430	1–2	2.19 @ 4430	2.19 @ 4340	0.07	2.22	2.09	-0.13	-0.13	**2.22**	2.10–2.26

Worked example – 12-1/4″ section: Min MW = 1.230 + 0.03 = 1.260 sg. Max static MW = 1.180 − 0.03 − 0.04 = 1.110 sg (weakest point in the loss zone at 2610 m). Window = -0.150 sg. Outside the loss zones the weakest FG is 1.544 sg at 1280 m, giving a window of +0.215 sg.

6.3 Interpretation

- **26″ – closed.** Some input tables assume 1.18 sg PP from about 100 m, while complete losses happened at 1.10–1.13 sg in the top 35 m (Balad-3, -4, -6). The shallow PP is probably overstated; top hole is drilled at minimum weight with losses expected.
- **17-1/2″ – open** (+0.22 sg).
- **12-1/4″ – open except the 2610–3210 m loss zone.** Drill at the minimum weight with LCM ready, which matches the field history (MW 1.17–1.40 with losses).
- **8-1/2″ – closed on paper.** The curves do not cross, but the 9-5/8″ shoe at 3596 m lies above the overpressure top (3680–4000 m), so the weakest open-hole FG (1.61 sg at 3630 m, just below the shoe) is far below the 2.23 sg needed deeper in the same section. Balad-1 and Balad-2 actually

drilled this interval at up to 2.25 sg, so either the 9-5/8″ was deeper in those wells or the real FG is higher than Eaton predicts. Per-well casing depths or a LOT at the shoe would settle it.

- **6″ – closed by -0.13 sg** (only 1–2 wells; low confidence).

“Recommended MW” is the minimum safe static mud weight rounded up to 0.01 sg. Where the window is closed it means: drill at the minimum, expect losses, and plan LCM or a contingency string.

7. Final chart and how to read it

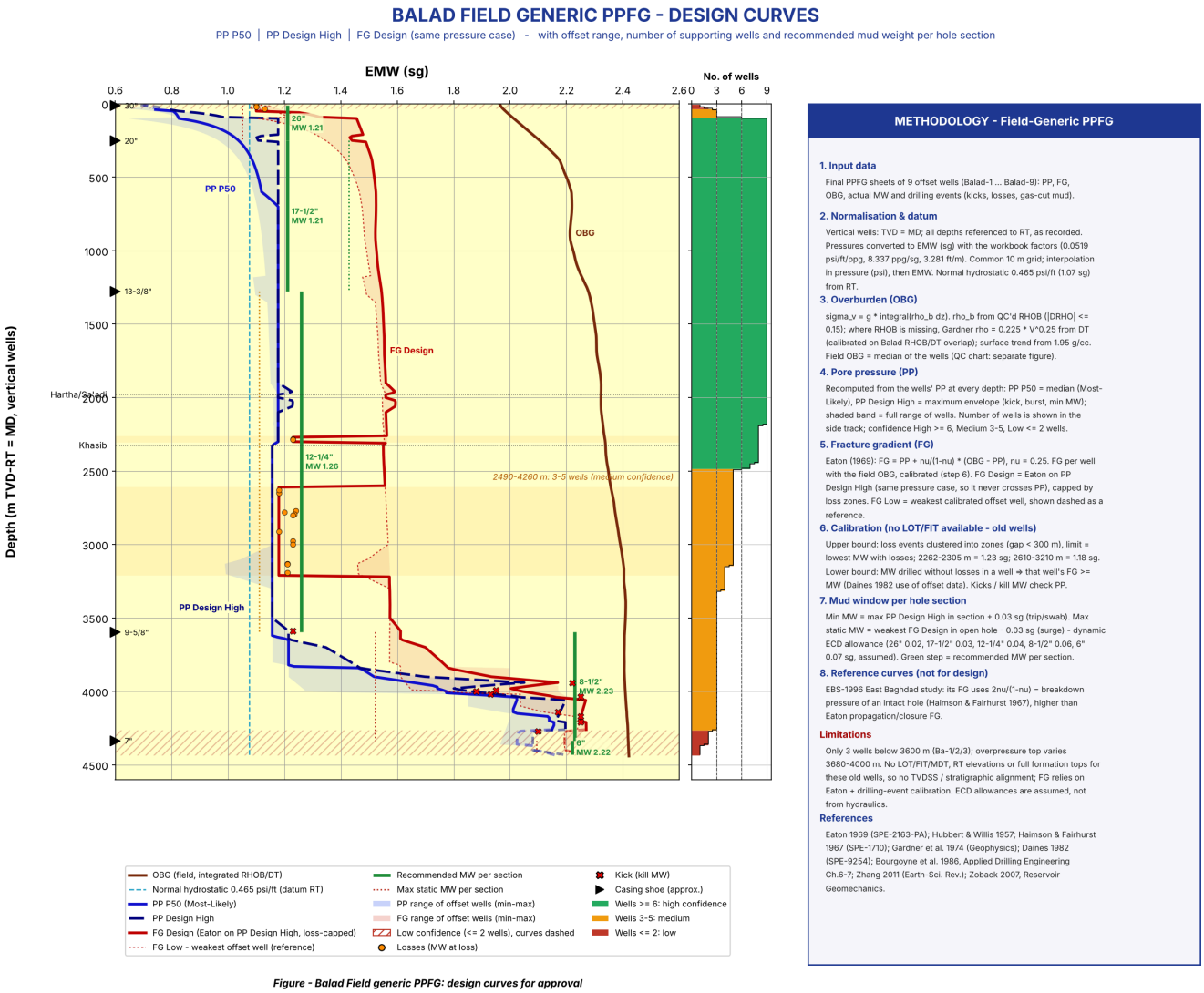


Figure - Balad Field generic PPFG: design curves for approval

Figure 2 – Balad field generic PPFG, design curves for approval.

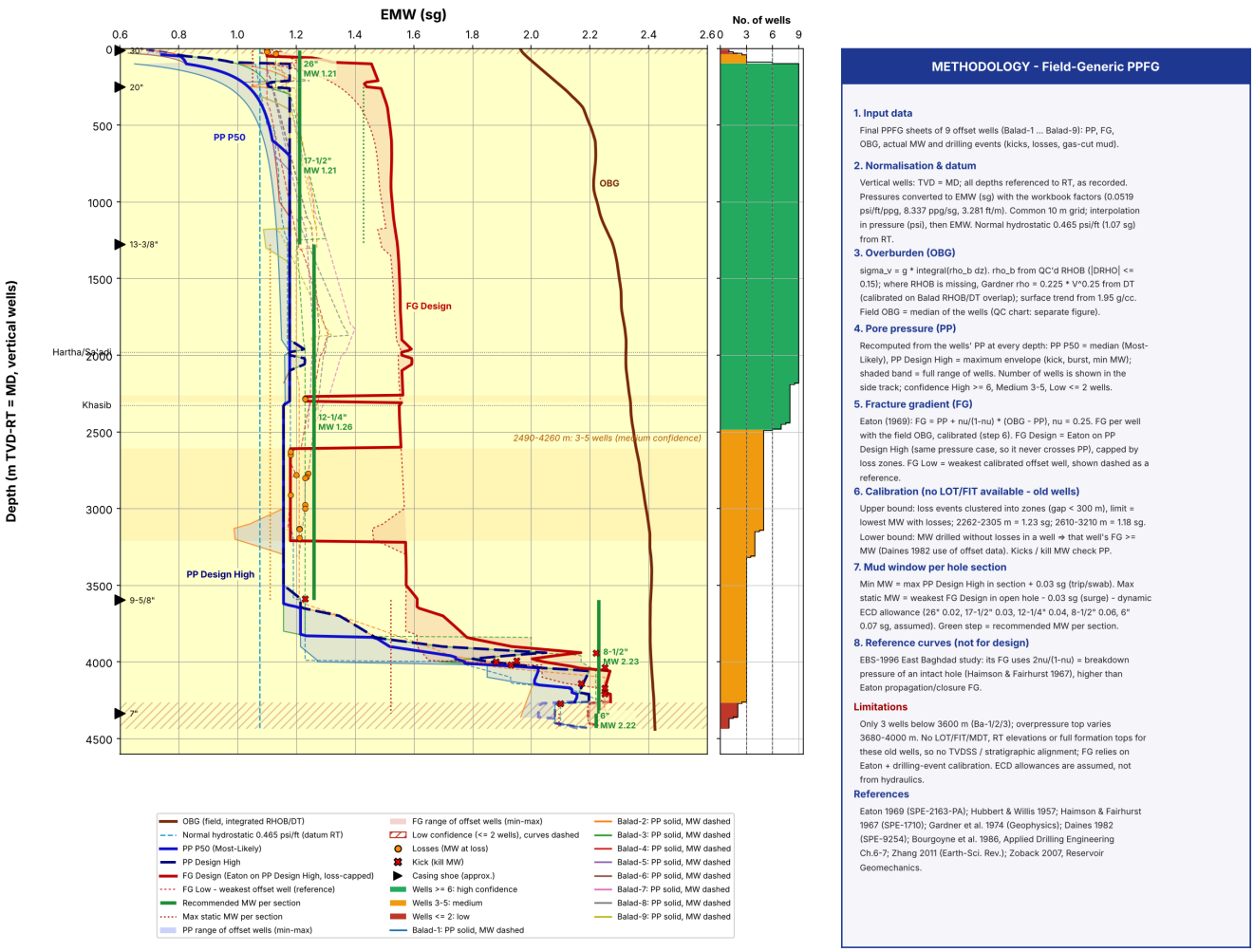


Figure - Balad Field generic PPFG: design curves for approval with the nine offset wells

Figure 2b – The same design curves with the nine offset wells (each well's PP solid, actual MW dashed).

Element	Meaning
Top axis / left axis	EMW in sg (0.6–2.6) / depth in m TVD-RT (= MD, vertical wells)
Brown curve	Field OBG integrated from RHOB/DT (Section 3)
Cyan dashed	Normal hydrostatic 0.465 psi/ft (1.07 sg) from RT
Blue solid / navy dashed	PP P50 / PP Design High (Section 4)
Red solid	FG Design – Eaton on PP Design High, capped by loss zones (Section 5)
Red dashed (thin)	FG Low – weakest calibrated offset well, reference only
Blue / red shaded bands	Min–max range of the wells' PP / calibrated FG
Green vertical bars + label	Recommended mud weight for each hole section (Section 6)
Dotted vertical lines	Max static MW per section: green open, orange open except loss zone, red closed
Orange dots / red crosses	Losses at the MW in use / kicks at the kill MW
Orange shaded bands	Loss zones
Hatched area, dashed curves	Low confidence: 2 wells or fewer
Right track	Number of wells at each depth: green ≥ 6, orange 3–5, red ≤ 2
Black triangles / dotted lines	Casing shoes (approx.) / formation tops

Using the curves for a new well

1. Read PP Design High and FG Design at the planned casing depths, and check the confidence track for those depths.
2. Use sheet *Design (ppg-psi)* of Balad_Unified_PPFG.xlsx as the PP/FG/OBG input to the casing-design software.
3. Use sheet *Hole-section window* for the planned mud weights, replacing the assumed ΔECD with hydraulics results.
4. Update the curves when new LOT/FIT, casing-depth or pressure data become available (re-run build_unified_ppfg.py).

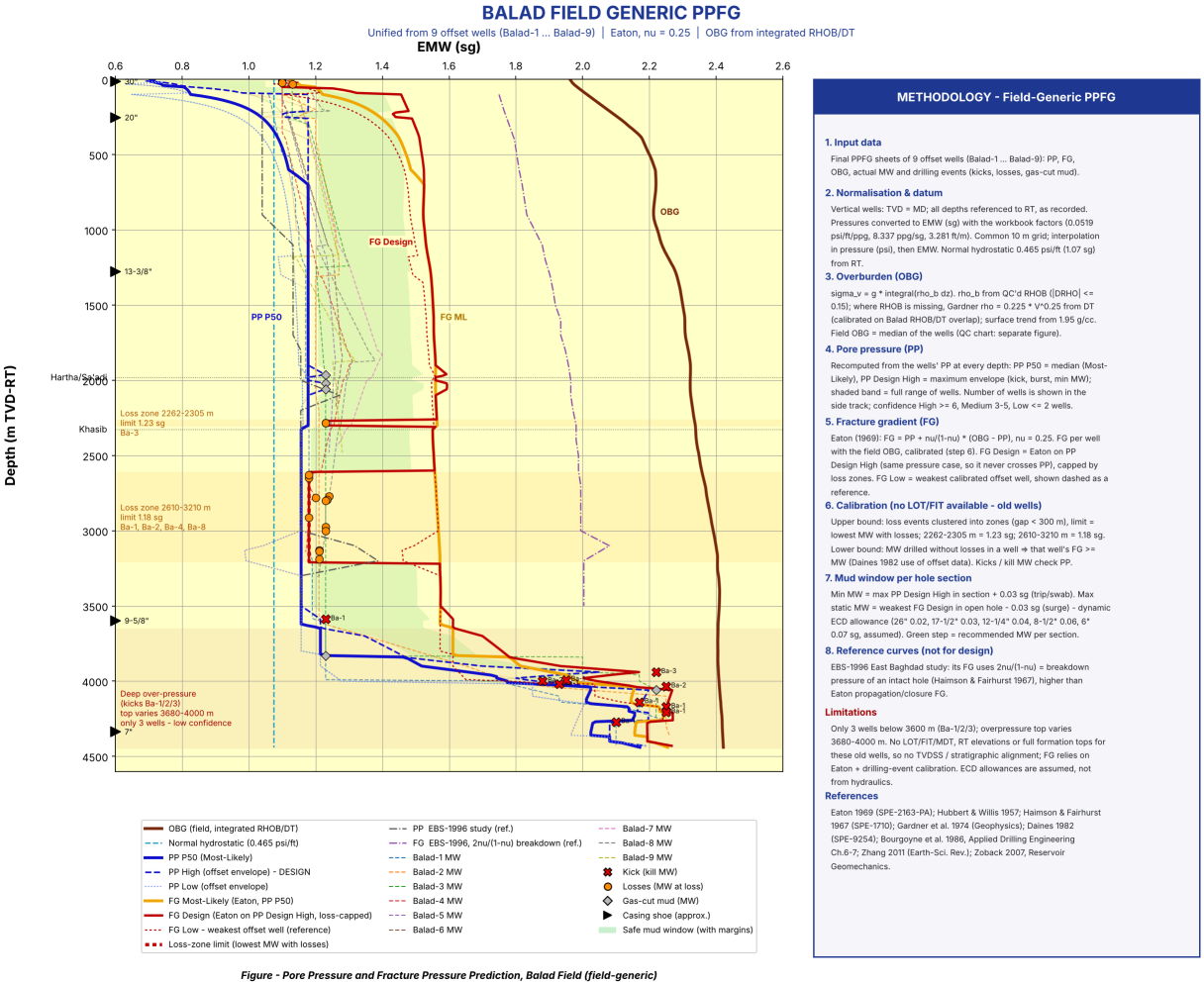


Figure 3 – QC chart: all wells' mud weights and PP, EBS-1996 reference curves and drilling events.

8. Limitations and data that would improve the result

Missing item	What it limits today	What it would change
LOT / FIT	FG rests on Eaton (ν = 0.25) and drilling-event bounds	Direct FG at each shoe; ν or stress ratio recalibrated, mainly below the 9-5/8″ shoe
Per-well casing depths	Casing points from one well; open-hole exposure per well unknown	Credit each well's open hole with the highest MW held; may reopen the 8-1/2″ window
MDT / RFT pressures	PP is interpreted, not measured	Measured points to anchor P50 and High
RT / GL elevations	Depths on each well's RT; no TVDSS	Common datum; hydrostatic from ground level
Formation tops for every well	Wells compared at equal depth, not equal stratigraphy	Stratigraphic alignment; narrows the 3680–4000 m overpressure-top spread
Hydraulics (ECD)	ΔECD 0.02–0.07 sg assumed	Real section windows

Wells per depth: 9 wells from 100 to 2190 m; fewer below; 3 wells below 3320 m (Balad-1, -2, -3); 2 below 4270 m; only Balad-1 below 4370 m. Values in the medium- and low-confidence intervals are provisional.

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Appendix A – Unified PPFG (every 50 m, EMW in sg)

Min MW = PP High + 0.03; Max MW = FG Design − 0.03 − ΔECD of the section. The full 10 m table is in sheet *Unified PPFG (sg)*.

Depth (m)	n	Conf.	OBG	PP Low	PP P50	PP High	FG ML	FG Low (weakest)	Loss limit	FG Design	Min MW	Max MW
50	3	Medium	1.98	0.74	0.80	0.83	1.20	1.15	1.10	1.10	0.86	1.05
100	9	High	2.01	0.65	0.83	1.18	1.22	1.10	–	1.46	1.21	1.41
150	9	High	2.04	0.83	0.94	1.18	1.31	1.23	–	1.47	1.21	1.42
200	9	High	2.07	0.91	1.00	1.18	1.36	1.30	–	1.48	1.21	1.43
250	9	High	2.10	0.97	1.04	1.11	1.39	1.35	–	1.44	1.14	1.39
300	9	High	2.13	1.00	1.06	1.18	1.42	1.38	–	1.50	1.21	1.44
350	9	High	2.16	1.03	1.08	1.18	1.44	1.40	–	1.51	1.21	1.45
400	9	High	2.18	1.05	1.09	1.18	1.45	1.42	–	1.51	1.21	1.45
450	9	High	2.19	1.06	1.10	1.18	1.46	1.44	–	1.51	1.21	1.45
500	9	High	2.20	1.07	1.11	1.18	1.47	1.45	–	1.52	1.21	1.46
550	9	High	2.21	1.08	1.11	1.18	1.48	1.46	–	1.52	1.21	1.46
600	9	High	2.22	1.09	1.12	1.18	1.49	1.47	–	1.52	1.21	1.46
650	9	High	2.22	1.10	1.15	1.18	1.51	1.47	–	1.52	1.21	1.46
700	9	High	2.22	1.10	1.18	1.18	1.52	1.47	–	1.52	1.21	1.46
750	9	High	2.22	1.11	1.18	1.18	1.52	1.48	–	1.52	1.21	1.46
800	9	High	2.22	1.11	1.18	1.18	1.52	1.48	–	1.52	1.21	1.46
850	9	High	2.21	1.12	1.18	1.18	1.52	1.48	–	1.52	1.21	1.46
900	9	High	2.21	1.12	1.18	1.18	1.52	1.48	–	1.52	1.21	1.46
950	9	High	2.22	1.12	1.18	1.18	1.52	1.49	–	1.52	1.21	1.46
1000	9	High	2.22	1.12	1.18	1.18	1.53	1.49	–	1.53	1.21	1.47
1050	9	High	2.23	1.13	1.18	1.18	1.53	1.49	–	1.53	1.21	1.47
1100	9	High	2.24	1.13	1.18	1.18	1.53	1.50	–	1.53	1.21	1.47
1150	9	High	2.25	1.13	1.18	1.18	1.53	1.50	–	1.53	1.21	1.47
1200	9	High	2.26	1.09	1.18	1.18	1.54	1.48	–	1.54	1.21	1.48
1250	9	High	2.27	1.09	1.18	1.18	1.54	1.49	–	1.54	1.21	1.48
1300	9	High	2.28	1.10	1.18	1.18	1.55	1.49	–	1.55	1.21	1.48
1350	9	High	2.29	1.14	1.18	1.18	1.55	1.52	–	1.55	1.21	1.48
1400	9	High	2.29	1.14	1.18	1.18	1.55	1.52	–	1.55	1.21	1.48
1450	9	High	2.30	1.14	1.18	1.18	1.55	1.53	–	1.55	1.21	1.48
1500	9	High	2.30	1.14	1.18	1.18	1.55	1.53	–	1.55	1.21	1.48
1550	9	High	2.30	1.14	1.18	1.18	1.55	1.53	–	1.55	1.21	1.48
1600	9	High	2.31	1.14	1.18	1.18	1.55	1.53	–	1.55	1.21	1.48
1650	9	High	2.31	1.15	1.18	1.18	1.55	1.53	–	1.55	1.21	1.48
1700	9	High	2.31	1.15	1.18	1.18	1.55	1.53	–	1.55	1.21	1.48
1750	9	High	2.31	1.15	1.18	1.18	1.56	1.54	–	1.56	1.21	1.49
1800	9	High	2.32	1.15	1.18	1.18	1.56	1.54	–	1.56	1.21	1.49
1850	9	High	2.32	1.15	1.18	1.18	1.56	1.54	–	1.56	1.21	1.49
1900	9	High	2.32	1.15	1.18	1.18	1.56	1.54	–	1.56	1.21	1.49
1950	9	High	2.32	1.17	1.18	1.22	1.56	1.55	–	1.59	1.25	1.52
2000	9	High	2.32	1.16	1.18	1.18	1.56	1.55	–	1.56	1.21	1.49
2050	9	High	2.32	1.16	1.18	1.23	1.56	1.55	–	1.59	1.26	1.52
2100	9	High	2.32	1.16	1.18	1.18	1.56	1.55	–	1.56	1.21	1.49
2150	9	High	2.33	1.16	1.18	1.18	1.56	1.55	–	1.56	1.21	1.49
2200	8	High	2.33	1.16	1.18	1.18	1.56	1.55	–	1.56	1.21	1.49
2250	8	High	2.33	1.16	1.18	1.18	1.56	1.55	–	1.56	1.21	1.49
2300	8	High	2.34	1.16	1.18	1.18	1.56	1.55	1.23	1.23	1.21	1.16
2350	8	High	2.34	1.16	1.16	1.16	1.55	1.55	–	1.55	1.19	1.48
2400	8	High	2.34	1.16	1.16	1.16	1.55	1.55	–	1.55	1.19	1.48
2450	7	High	2.34	1.16	1.16	1.16	1.55	1.55	–	1.55	1.19	1.48
2500	5	Medium	2.35	1.16	1.16	1.16	1.55	1.55	–	1.55	1.19	1.48
2550	5	Medium	2.35	1.16	1.16	1.16	1.56	1.56	–	1.56	1.19	1.49
2600	5	Medium	2.36	1.16	1.16	1.16	1.56	1.56	–	1.56	1.19	1.49
2650	5	Medium	2.36	1.16	1.16	1.16	1.56	1.56	1.18	1.18	1.19	1.11
2700	5	Medium	2.37	1.16	1.16	1.16	1.56	1.56	1.18	1.18	1.19	1.11
2750	5	Medium	2.37	1.16	1.16	1.16	1.56	1.56	1.18	1.18	1.19	1.11
2800	5	Medium	2.37	1.16	1.16	1.16	1.56	1.56	1.18	1.18	1.19	1.11
2850	5	Medium	2.38	1.16	1.16	1.16	1.56	1.56	1.18	1.18	1.19	1.11

Depth (m)	n	Conf.	OBG	PP Low	PP P50	PP High	FG ML	FG Low (weakest)	Loss limit	FG Design	Min MW	Max MW
2900	5	Medium	2.38	1.16	1.16	1.16	1.57	1.57	1.18	1.18	1.19	1.11
2950	5	Medium	2.39	1.16	1.16	1.16	1.57	1.57	1.18	1.18	1.19	1.11
3000	5	Medium	2.39	1.16	1.16	1.16	1.57	1.57	1.18	1.18	1.19	1.11
3050	5	Medium	2.40	1.10	1.16	1.16	1.57	1.53	1.18	1.18	1.19	1.11
3100	5	Medium	2.40	1.04	1.16	1.16	1.57	1.50	1.18	1.18	1.19	1.11
3150	4	Medium	2.40	0.99	1.16	1.16	1.57	1.46	1.18	1.18	1.19	1.11
3200	4	Medium	2.40	0.99	1.16	1.16	1.57	1.46	1.18	1.18	1.19	1.11
3250	4	Medium	2.40	1.08	1.16	1.16	1.57	1.52	–	1.57	1.19	1.50
3300	4	Medium	2.41	1.16	1.16	1.16	1.57	1.57	–	1.57	1.19	1.50
3350	3	Medium	2.41	1.16	1.16	1.16	1.57	1.57	–	1.57	1.19	1.50
3400	3	Medium	2.41	1.16	1.16	1.16	1.57	1.57	–	1.57	1.19	1.50
3450	3	Medium	2.41	1.16	1.16	1.16	1.57	1.57	–	1.57	1.19	1.50
3500	3	Medium	2.41	1.16	1.16	1.16	1.57	1.57	–	1.57	1.19	1.50
3550	3	Medium	2.41	1.16	1.16	1.19	1.57	1.57	–	1.60	1.22	1.53
3600	3	Medium	2.41	1.16	1.16	1.21	1.57	1.57	–	1.61	1.24	1.52
3650	3	Medium	2.41	1.16	1.21	1.22	1.61	1.57	–	1.62	1.25	1.53
3700	3	Medium	2.41	1.16	1.21	1.35	1.61	1.57	–	1.70	1.38	1.61
3750	3	Medium	2.41	1.16	1.21	1.39	1.61	1.57	–	1.73	1.42	1.64
3800	3	Medium	2.41	1.16	1.21	1.43	1.61	1.57	–	1.76	1.46	1.67
3850	3	Medium	2.41	1.21	1.48	1.50	1.79	1.61	–	1.80	1.53	1.71
3900	3	Medium	2.41	1.21	1.52	1.70	1.81	1.61	–	1.93	1.73	1.84
3950	3	Medium	2.41	1.25	1.69	1.98	1.93	1.63	–	2.12	2.01	2.03
4000	3	Medium	2.41	1.27	1.77	1.86	1.99	1.78	–	2.04	1.89	1.95
4050	3	Medium	2.41	1.85	2.02	2.13	2.15	2.04	–	2.22	2.16	2.13
4100	3	Medium	2.41	1.85	2.02	2.19	2.15	2.04	–	2.26	2.22	2.17
4150	3	Medium	2.41	2.01	2.03	2.17	2.15	2.15	–	2.25	2.20	2.16
4200	3	Medium	2.42	2.00	2.14	2.16	2.23	2.23	–	2.25	2.19	2.16
4250	3	Medium	2.42	1.99	2.15	2.20	2.24	2.24	–	2.27	2.23	2.18
4300	2	Low	2.42	1.98	2.03	2.08	2.16	2.19	–	2.19	2.11	2.10
4350	2	Low	2.42	1.97	2.02	2.08	2.16	2.19	–	2.19	2.11	2.09
4400	1	Low	2.42	2.08	2.08	2.08	2.19	2.19	–	2.19	2.11	2.09

Appendix B – Per-well PP and calibrated FG (every 100 m, sg)

Depth (m)	PP Ba-1	PP Ba-2	PP Ba-3	PP Ba-4	PP Ba-5	PP Ba-6	PP Ba-7	PP Ba-8	PP Ba-9	FG Ba-1	FG Ba-2	FG Ba-3	FG Ba-4	FG Ba-5	FG Ba-6	FG Ba-7	FG Ba-8	FG Ba-9
100	0.65	1.00	1.00	0.83	0.83	0.83	0.83	1.18	0.83	1.10	1.34	1.34	1.22	1.22	1.22	1.22	1.46	1.22
200	0.91	1.18	1.09	1.00	1.00	1.00	1.00	1.18	1.00	1.30	1.48	1.42	1.36	1.36	1.36	1.36	1.48	1.36
300	1.00	1.18	1.18	1.06	1.06	1.06	1.06	1.06	1.06	1.38	1.50	1.50	1.42	1.42	1.42	1.42	1.42	1.42
400	1.05	1.18	1.18	1.09	1.09	1.09	1.18	1.09	1.18	1.42	1.51	1.51	1.45	1.45	1.45	1.51	1.45	1.51
500	1.07	1.18	1.18	1.11	1.11	1.11	1.18	1.11	1.18	1.45	1.52	1.52	1.47	1.47	1.47	1.52	1.47	1.52
600	1.09	1.18	1.18	1.12	1.12	1.12	1.18	1.12	1.18	1.47	1.52	1.52	1.49	1.49	1.49	1.52	1.49	1.52
700	1.10	1.18	1.18	1.13	1.13	1.18	1.18	1.18	1.18	1.47	1.52	1.52	1.49	1.49	1.52	1.52	1.52	1.52
800	1.11	1.18	1.18	1.13	1.18	1.18	1.18	1.18	1.18	1.48	1.52	1.52	1.49	1.52	1.52	1.52	1.52	1.52
900	1.12	1.18	1.18	1.14	1.18	1.18	1.18	1.18	1.18	1.48	1.52	1.52	1.50	1.52	1.52	1.52	1.52	1.52
1000	1.12	1.18	1.18	1.14	1.18	1.18	1.18	1.18	1.18	1.49	1.53	1.53	1.50	1.53	1.53	1.53	1.53	1.53
1100	1.13	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.50	1.53	1.53	1.53	1.53	1.53	1.53	1.53	1.53
1200	1.13	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.09	1.51	1.54	1.54	1.54	1.54	1.54	1.54	1.54	1.48
1300	1.14	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.10	1.52	1.55	1.55	1.55	1.55	1.55	1.55	1.55	1.49
1400	1.14	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.52	1.55	1.55	1.55	1.55	1.55	1.55	1.55	1.55
1500	1.14	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.53	1.55	1.55	1.55	1.55	1.55	1.55	1.55	1.55
1600	1.14	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.53	1.55	1.55	1.55	1.55	1.55	1.55	1.55	1.55
1700	1.15	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.53	1.55	1.55	1.55	1.55	1.55	1.55	1.55	1.55
1800	1.15	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.54	1.56	1.56	1.56	1.56	1.56	1.56	1.56	1.56
1900	1.15	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.54	1.56	1.56	1.56	1.56	1.56	1.56	1.56	1.56
2000	1.16	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.55	1.56	1.56	1.56	1.56	1.56	1.56	1.56	1.56
2100	1.16	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.18	1.55	1.56	1.56	1.56	1.56	1.56	1.56	1.56	1.56
2200	1.16	1.18	1.18	1.18	1.18	–	1.18	1.18	1.18	1.55	1.56	1.56	1.56	1.56	–	1.56	1.56	1.56
2300	1.16	1.18	1.18	1.18	1.18	–	1.18	1.18	1.18	1.55	1.56	1.56	1.56	1.56	–	1.56	1.56	1.56
2400	1.16	1.16	1.16	1.16	1.16	–	1.16	1.16	1.16	1.55	1.55	1.55	1.55	1.55	–	1.55	1.55	1.55
2500	1.16	1.16	1.16	1.16	–	–	–	1.16	–	1.55	1.55	1.55	1.55	–	–	–	1.55	–
2600	1.16	1.16	1.16	1.16	–	–	–	1.16	–	1.56	1.56	1.56	1.56	–	–	–	1.56	–
2700	1.16	1.16	1.16	1.16	–	–	–	1.16	–	1.56	1.56	1.56	1.56	–	–	–	1.56	–
2800	1.16	1.16	1.16	1.16	–	–	–	1.16	–	1.56	1.56	1.56	1.56	–	–	–	1.56	–
2900	1.16	1.16	1.16	1.16	–	–	–	1.16	–	1.57	1.57	1.57	1.57	–	–	–	1.57	–
3000	1.16	1.16	1.16	1.16	–	–	–	1.16	–	1.57	1.57	1.57	1.57	–	–	–	1.57	–
3100	1.16	1.04	1.16	1.16	–	–	–	1.16	–	1.57	1.50	1.57	1.57	–	–	–	1.57	–
3200	1.16	0.99	1.16	–	–	–	–	1.16	–	1.57	1.46	1.57	–	–	–	–	1.57	–
3300	1.16	1.16	1.16	–	–	–	–	1.16	–	1.57	1.57	1.57	–	–	–	–	1.57	–
3400	1.16	1.16	1.16	–	–	–	–	–	–	1.57	1.57	1.57	–	–	–	–	–	–
3500	1.16	1.16	1.16	–	–	–	–	–	–	1.57	1.57	1.57	–	–	–	–	–	–
3600	1.21	1.16	1.16	–	–	–	–	–	–	1.61	1.57	1.57	–	–	–	–	–	–
3700	1.21	1.35	1.16	–	–	–	–	–	–	1.61	1.70	1.57	–	–	–	–	–	–
3800	1.21	1.43	1.16	–	–	–	–	–	–	1.61	1.76	1.57	–	–	–	–	–	–
3900	1.21	1.52	1.70	–	–	–	–	–	–	1.61	1.81	2.00	–	–	–	–	–	–
4000	1.27	1.86	1.68	–	–	–	–	–	–	1.78	2.04	2.17	–	–	–	–	–	–
4100	1.85	2.02	2.19	–	–	–	–	–	–	2.04	2.25	2.26	–	–	–	–	–	–
4200	2.14	2.00	2.16	–	–	–	–	–	–	2.23	2.25	2.25	–	–	–	–	–	–
4300	2.08	1.98	–	–	–	–	–	–	–	2.19	2.25	–	–	–	–	–	–	–
4400	2.08	–	–	–	–	–	–	–	–	2.19	–	–	–	–	–	–	–	–

Appendix C – Drilling events used

Well	Depth (m)	Type	MW (sg)	Kill / max MW (sg)	Note (as recorded)
Balad-1	2650	Losses	1.18	–	Partial Losses (solved with LCM, no change to density)
Balad-1	3588	Kick	1.19	1.23	Kick (MW dropped from 1.19 to 1.0, killed with 1.23)
Balad-1	3991	Kick	1.23	1.95	Kick (Circulated with 1.28, killed with 1.9-1.95, resumed with 1.78)
Balad-1	4020	Kick	1.78	1.93	Kick (Killed with 1.85, resumed with 1.93)
Balad-1	4139	Kick	1.93	2.17	Kick (Killed with 2.17, resumed with 2.11)
Balad-1	4171	Kick	2.11	2.25	Kick (Killed and resumed with 2.25)
Balad-1	4206	Kick	2.17	2.25	Kick (Killed and resumed with 2.25)
Balad-1	4271	Kick	2.10	–	Kick (Killed and resumed with 2.1)
Balad-2	3130	Losses	1.21	–	Compete Losses
Balad-2	3135	Losses	1.21	–	Partial Losses (15m3/hr)
Balad-2	3190	Losses	1.21	–	Partial Losses
Balad-2	4000	Kick	1.78	1.88	Kick (Killed and resumed with 1.88)
Balad-2	4035	Kick	1.88	2.25	Kick (Killed with 2.2 and resumed with 2.25)
Balad-3	33	Losses	1.13	–	Severe mud loss with hole completely collapsed.
Balad-3	1964	Gas-cut mud	1.23	–	Gas cut mud reduced density to 1.13 from 1.23 and circulated to normal density
Balad-3	2021	Gas-cut mud	1.23	–	Mud density dropped to 1.08 and controlled by normal circulation
Balad-3	2059	Gas-cut mud	1.23	–	Mud density dropped to 1.08 and controlled by normal circulation
Balad-3	2282	Losses	1.23	–	Complete mud losses encountered
Balad-3	2285	Losses	1.23	–	Complete mud losses encountered
Balad-3	3832	Gas-cut mud	1.23	1.24	Mud density dropped to 1.05 from 1.23 due to gas cut mud and killed with 1.24
Balad-3	3940	Kick	2.00	2.22	Well kicked during coring and killed with 2.22, then losses occured during killing then reduced to 2.17
Balad-3	4058	Gas-cut mud	2.22	–	Gas cut mud from 4058 to 4260
Balad-4	2	Losses	1.10	–	Complete losses encountered
Balad-4	15	Losses	1.10	–	Complete losses encountered
Balad-4	2630	Losses	1.18	–	Partial Losses
Balad-4	2782	Losses	1.20	–	Partial Losses
Balad-4	2912	Losses	1.18	–	Partial Losses
Balad-6	21	Losses	1.10	–	Complete Mud Losses
Balad-8	2770	Losses	1.24	–	Partial Losses treated with LCM
Balad-8	2793	Losses	1.24	–	Partial Losses treated with LCM
Balad-8	2800	Losses	1.23	–	Partial Losses treated with LCM
Balad-8	2975	Losses	1.23	–	Partial Losses treated with LCM
Balad-8	3000	Losses	1.23	–	Partial Losses treated with LCM